ALGEBRAIC NUMBER THEORY- SHEET 8

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Solutions to Exercise 8.2 should be uploaded to moodle or handed to me, or put in the homework box, by: 17:00 on 25/03/2022.

Exercise 8.1. (1) Let $K = \mathbb{Q}(\sqrt{-13})$. Compute the class group Cl_K and give the multiplication table.
(2) Find all integer solutions to the Diophantine equation $x^3 = y^2 + 13$, justifying your answers carefully.

Exercise 8.2. (1) Let $K = \mathbb{Q}(\sqrt{-21})$. Compute the class group Cl_K and give the multiplication table.
(2) Find all integer solutions to the Diophantine equation $x^3 = y^2 + 21$, justifying your answers carefully.

Exercise 8.3. (1) Let $K = \mathbb{Q}(\sqrt{-17})$. Compute the class group Cl_K and give the multiplication table.
(2) Find all integer solutions to the Diophantine equation $x^3 = y^2 + 17$, justifying your answers carefully.

Exercise 8.4. (1) Let $K = \mathbb{Q}(\sqrt{-79})$. Compute the class group Cl_K and give the multiplication table.
(2) Find all integer solutions to the Diophantine equation $x^3 = y^2 - y + 20$, justifying your answers carefully.

Exercise 8.5. Let ζ_7 be a 7-th root of unity and $K = \mathbb{Q}(\zeta_7)$. Complete the following table describing the decomposition of ideals (p) (with p a prime number) in $\mathcal{O}_K$ as done in Example 3.6.16 of the notes. In the table, we let $n = p^k m$ where $p \nmid m$.

$p \mod 7$	Order of $p \mod m$	Factorization of (p)	Norms
0			
1			
2			
3			
4			
5			
6			